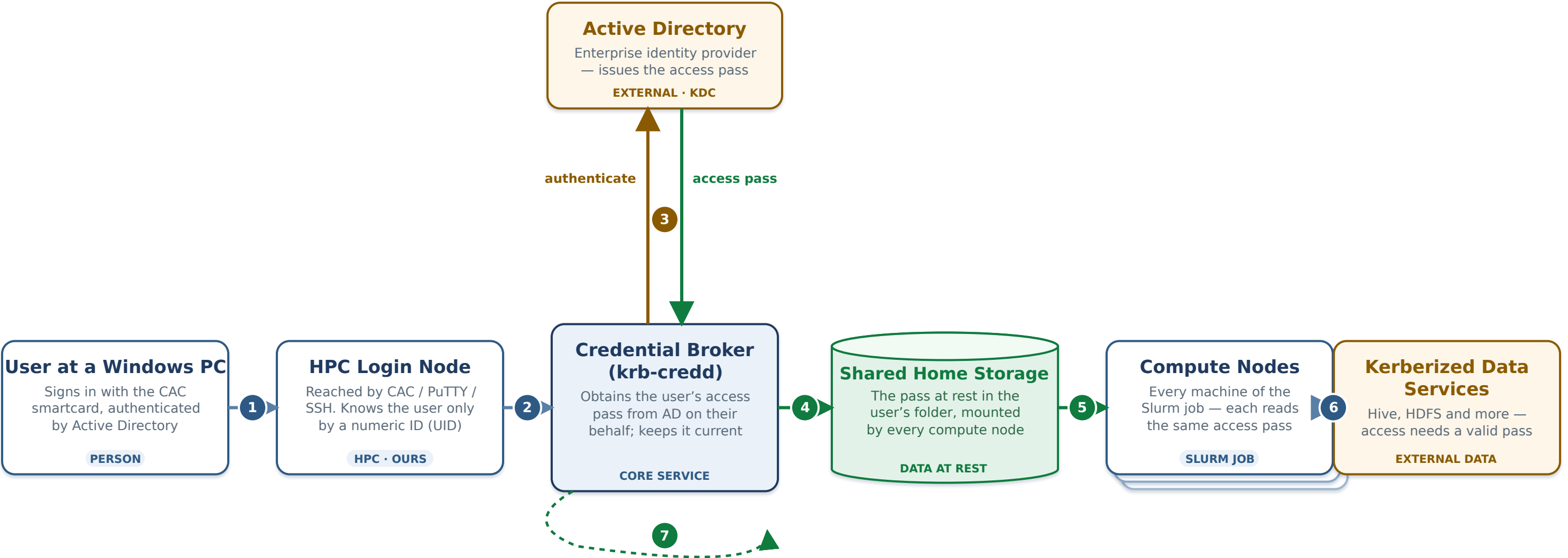


Secure Data Access for HPC Jobs — Executive Data Flow

How a user’s identity becomes a short-lived access pass that follows their job to every machine and every data service



How it works — the steps

- 1 User signs in to the login node with their CAC (as they do today).
- 2 The login node asks the Credential Broker for that user’s pass.
- 3 The Broker authenticates to Active Directory and receives the pass.
- 4 The Broker writes the pass into the user’s home folder, as the user.
- 5 Because home is shared, every compute node already sees the pass.
- 6 Each task shows the pass to the data services and reads its data.
- 7 The Broker keeps the pass fresh for as long as the job runs.

What flows across every arrow

One thing moves through the system: the user’s **access pass** — a short-lived Kerberos ticket that proves who they are. It is never a password, is created only after authentication, and expires on its own.

- Person / external data
- HPC service (ours)
- Data at rest
- Access-pass flow

Takeaway: users keep using their CAC; each job automatically gets a temporary, self-expiring pass — no stored passwords, no manual steps.